

zero_shortcomings_audit

scripts/

zero_shortcomings_audit.sh

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Purpose: Unified enumeration, closure-evidence re-verification, and standing-check entry point across this project's three tracked "unfinished/gap/shortcoming" surfaces (open backlog items, unimplemented named gates, open coverage-escapes). See specs/003-zero-shortcomings-audit/contracts/cli.md for the full CLI contract.

Usage: scripts/zero_shortcomings_audit.sh <enumerate|verify-closure|standing-check> [options]

-h / --help prints the usage text and exits 0. Running with no mode prints the usage and exits 1. An unknown mode prints unknown mode: <x> plus the usage and exits 1. (The built-in --help text lists every mode and option; this guide gives the full semantics.)

Modes

enumerate [--json] [--surface S] [--sort-by-risk]

Read-only. S is one of all (default), backlog, gates, escapes, blocked.

- Default text output: Open backlog items, Unimplemented gates, Open coverage-escapes counts (only for the selected surface).
- --json: one JSON object with the keys backlog_open, gates_unimplemented, escapes_open (only the selected surfaces' keys).
- --surface blocked: lists every Operator-blocked item as id|unblock_condition (joined from operator_block_details), never a bare count.

- `--sort-by-risk` (FR-012): prints the open backlog item ids ordered by reopen count descending (count of Reopened events in `item_history`), then `last_modified` descending, then `id`. With `--json` it prints a JSON array of ids. **Only valid with `--surface backlog`.**
- Errors (all exit 1, message on `stderr`): unknown option (enumerate: unknown option: X); unrecognized `--surface` value (enumerate: unrecognized `--surface` value: 'X' (expected `all|backlog|gates|escapes|blocked`)); `--sort-by-risk` with any surface other than `backlog` (enumerate: `--sort-by-risk` applies only to `--surface backlog` (got 'X')).
- The gate count fails loud (exit 1, message naming the gate script) when the gate-ledger script prints no parseable `unimplemented=N` line, rather than reporting 0.

`verify-closure <item-id> [--reopen-on-mismatch] [--require-layer L]`

Independently re-runs the recorded closure evidence for one item. Evidence is read from the first `closure_evidence_*.md` directly inside `<AUDIT_QA_ROOT>/<item-id>/`. `--require-layer L` (`source|artifact|runtime`, default `runtime`) is the minimum evidence layer accepted; the evidence's `**Evidence Layer:**` field (default `source` when absent) must rank at or above it (`source < artifact < runtime`; an unrecognized label ranks as `source`).

Required evidence-file fields:

- `**Evidence Layer:**` (optional; see above)
- `**Test Type:**` (FR-010) – required. Allowed values, compared case-insensitively after trimming: `unit`, `integration`, `e2e`, `security`, `stress`, `chaos`, `scaling`, `ui`, `challenge`.
- `**Command:**` ``<command>`` – the command to re-run (evaluated with `eval`).
- `**Result Summary:**` `<text>` – compared exactly against the command's fresh `stdout`.

Exit codes:

Code	Meaning
0	The recorded command's fresh output equals the recorded Result Summary (reproduced its recorded evidence).

Code	Meaning
1	Mismatch (MISMATCH -- recorded '...', got '...', values credential-redacted before printing); or usage error: missing item id, unknown option.
2	No closure_evidence_*.md found; declared layer below --require-layer; no **Test Type:**; invalid **Test Type:**; no **Command:** field.

--reopen-on-mismatch: on a mismatch, calls the workable-items reopen CLI (--why test-failed --who AI, evidence path = the evidence file) before returning 1.

Evidence-corruption guard (FR-008): before running the recorded command the script snapshots (content hashes) every git-tracked file under docs/qa/; afterwards any such file changed **outside** <AUDIT_QA_ROOT>/<item-id>/ is restored (files that were already dirty before the run are restored from a byte copy of their pre-command content, others via git checkout --) and the incident is appended to docs/qa/zero_shortcomings_audit/<run-id>.log with a warning on the console. The guard never changes the exit code by itself.

standing-check

Recurring, advisory mode; **always exits 0**. Runs enumerate --json for each of the backlog, gates, escapes surfaces and appends one line to <AUDIT_STANDING_LOG_DIR>/<run-id>.log:

```
<run-id> mode=standing-check status=ok
{"backlog_open":N,"gates_unimplemented":N,"escapes_open":N}
```

If a sub-check fails, the line instead carries status=degraded reason=<list> failed=<list> (reasons such as db_missing, <surface>_rc<N>) and a WARN is printed; the JSON body then holds only the surfaces that succeeded. A missing/unreadable docs/workable_items.db is detected before any sqlite3 call (so a missing file is never silently created as an empty DB) and recorded as db_missing. If redaction itself fails the line falls back to status=degraded reason=redaction_failed.

Run id format: YYYYMMDDTHHMMSSZ-pid<PID>.

Environment overrides

Variable	Default	Effect
WORKABLE_ITEMS_DB_OVERRIDE	docs/workable_items.db	Backlog / blocked / reopen SQLite DB path.
GATE_LEDGER_SCRIPT_OVERRIDE	constitution/scripts/ gates/ cm_gate_ledger_ratchet.sh	Gate-ledger script used for the unimplemented-gate count.
AUDIT_QA_ROOT	docs/qa	Root searched for <item-id>/closure_evidence_*.md.
AUDIT_STANDING_LOG_DIR	docs/qa/ zero_shortcomings_audit	Directory for the standing-check log (created if missing).

Redaction (Constitution Principle III / §11.4.10)

Every value printed on a mismatch, every command quoted in a corruption incident, and every standing-check log line passes through `audit_redact_before_write`: the value half of a keyword<sep>value pair (sep = : or =) is replaced by <redacted-per-§11.4.10> while the keyword is kept. Keywords (case-insensitive): password, passwd, secret, api_key (also api-key/apiskey), access_token, auth_token, client_secret, plus the project extension key, token, cookies. Comparison itself uses the raw values; only what is printed or written is redacted. Bare key/token over-redact by design.

Inputs: docs/workable_items.db (read-only), constitution/scripts/gates/ (read-only, via cm_gate_ledger_ratchet.sh), docs/QA_DISCOVERY_LEDGER.md (read-only), docs/qa/<item-id>/closure_evidence_*.md (read; verify-closure).

Outputs: A human-readable or `--json` report; `standing-check` appends one line to `docs/qa/zero_shortcomings_audit/<run-id>.log`; `verify-closure` appends to the same log only when the corruption guard reverted something.

Side-effects: None in `enumerate`. `standing-check` writes its log line. `verify-closure` executes the recorded command (`eval`), may revert corrupted tracked evidence files, and with `--reopen-on-mismatch` may call the `workable-items` CLI to reopen an item.

Dependencies: `sqlite3`, `git`, `constitution/scripts/workable-items/workable-items`, `constitution/scripts/gates/cm_gate_ledger_ratchet.sh`, and `scripts/lib/audit_*.sh`.

Cross-references: `specs/003-zero-shortcomings-audit/{spec,plan,research,data-model}.md`.

Last verified: 2026-09-26 (against the script source; the `--help` text itself was run).